

H&E-first molecular triage for sarcoma: a reproducible proof of concept on open data

AI in Orthopaedics and MSK 2026 Hackathon, submission document

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Executive summary

Problem. Sarcoma subtype decides the operation and the oncology plan, and many subtypes are defined by a gene fusion. The molecular test that names the fusion is slow and centralised: in England urgent solid tumour panels have a 21-day target ([East Genomics](#)), in Taiwan National Health Insurance reimburses next-generation sequencing once per lifetime and only at regional hospitals or above ([Health Promotion Administration](#)).

Solution. A model that reads the routine H&E slide already produced for every biopsy and returns, in minutes, a subtype probability and a recommendation for which genomic test to order first. It does not replace the test. It makes the first order the right one.

Efficiency lever. One H&E slide plus one model call replaces up to three weeks of waiting and one avoidable round of testing per misrouted case.

Result so far. A reproducible baseline on open TCGA-SARC data separates two common subtypes from slide overviews with an out-of-fold AUC of 0.82 on 60 patients, and runs live in the browser at <https://iressa-sarcoma-triage-demo.static.hf.space/>.

What is in this submission.

File	Content
technical/01_sarcoma_hne_triage.ipynb	Working baseline on open TCGA-SARC data, 60 patients, out-of-fold AUC 0.82
technical/technical_report.md	Section-by-section write-up of how the baseline model was built, trained and evaluated (Technical report, sections 1 to 10)
technical/02_savings_model.ipynb	Transparent savings arithmetic, every input labelled sourced or assumption
medicine/clinical_implementation.md	Medical part: the clinical problem, where the AI sits, integration and safety, Taiwan, diversity, implementation stages
business/business_case.md	Business part on the NHS five-case model: strategic, economic, commercial, financial, management, including business model, competition, milestones, funding and patient involvement
technical/requirements.txt, technical/fetch_overviews.py group-member-contact/README.md	Environment and download helper Team and contributions

Medical part: clinical implementation strategy

The clinical problem, where the AI sits in the pathway, how it is integrated and kept safe, the Taiwan setting, diversity and equity, and the stages of clinical implementation.

1. The clinical problem

Why the subtype matters. A sarcoma is not one disease (Figure 1). More than 100 bone and soft tissue subtypes exist, and many are defined by a single pathognomonic gene fusion rather than by what the tumour looks like, for

Sarcoma is not one disease

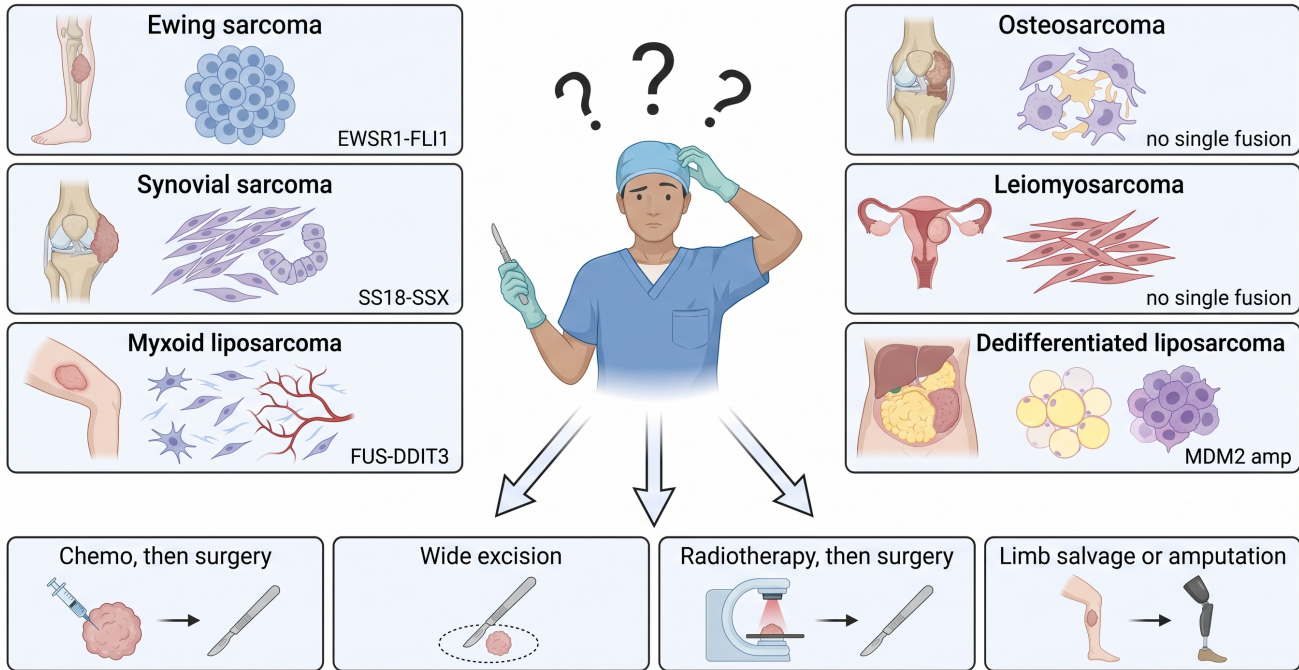


Figure 1: Sarcoma is not one disease: six subtypes with six genetic definitions, one surgeon who has to choose between four different treatment pathways

example Ewing sarcoma, synovial sarcoma and myxoid liposarcoma ([Sbaraglia 2021](#)). So the subtype is not a label added at the end. It is the first thing the surgical team has to know, because every later decision depends on it.

What the subtype decides for the surgeon. Three orthopaedic decisions follow directly from the subtype. First, whether to operate now or treat first: Ewing sarcoma and high-grade osteosarcoma receive several cycles of chemotherapy before any resection, whereas most adult soft tissue sarcomas go to surgery first ([ESMO bone sarcoma guideline](#), [Strauss 2021](#), [UK bone sarcoma guidelines 2024](#)). Second, how wide to cut and whether the limb can be saved: the planned margin, the need for pre-operative radiotherapy, and the choice between limb-salvage reconstruction and amputation are all set by subtype and grade ([UK soft tissue sarcoma guidelines 2024](#)). Third, who should operate and where: in England every suspected bone sarcoma must be referred to a designated centre such as the Royal National Orthopaedic Hospital, which with UCLH sees about 500 new sarcoma patients a year ([RNOH sarcoma unit](#)). Operating on the wrong subtype means the wrong margin, the wrong sequence, or an avoidable amputation.

The diagnostic pathway today:

1. A lump is imaged and biopsied at a local hospital.
2. The H&E slide is read by a general pathologist, who suspects sarcoma.
3. The case is referred to one of the designated sarcoma centres. All suspected bone sarcomas in England must go to such a centre ([RNOH sarcoma unit](#)).
4. The specialist pathologist orders immunohistochemistry and a genomic test through the Genomic Laboratory Hub. The national target for urgent solid tumour panels is 90 per cent reported within 21 days ([East Genomics](#)).
5. The multidisciplinary team meets when the molecular result is back.

Where the time goes: steps 3 and 4. A misrouted or repeated genomic request adds a full cycle.

Why the subtype is an efficiency problem. The subtype is confirmed by a molecular test, and that test is slow and centralised. Figure 3 puts the three points side by side. Here is what it costs when the test is wrong or late, with the numbers.

- **England, the test.** Sarcoma genomic testing runs through seven Genomic Laboratory Hubs, with 54 sarcoma indications in the National Genomic Test Directory ([Birmingham Women's and Children's, genomic testing in](#)

sarcomas). The national standard for urgent solid tumour panels is 90 per cent reported within 21 days ([East Genomics, turnaround times](#)). The target is 90 per cent of tests reported within 21 days. In practice about one test in three is late: the only hub that publishes its figure delivers “over two thirds” on time and calls that “slightly above the national average” ([South East Genomics, turnaround times](#)), and NHS England publishes no national figure at all ([Future Health, Keeping pace with cancer, 2024](#)). So a request sent to the wrong panel is not a short delay. It is a second three-week wait at the back of a queue that is already running late.

- **England, the patient.** The delay is not only in the genetic test. The whole cancer pathway in England is already slower than its own target. In March 2026 only 72.8 per cent of patients began their first treatment within 62 days of an urgent referral; the target is 85 per cent ([NHS England Cancer Waiting Times, March 2026](#)). For sarcoma specifically, Sarcoma UK’s National Sarcoma Survey found that 30 per cent of patients waited at least six months after first consulting a healthcare professional before an accurate diagnosis, 17 per cent waited more than a year, and 35 per cent saw a healthcare professional more than three times before being referred for tests ([Sarcoma UK, Delays cost lives, 2020](#)). The same report records that delay is associated with higher risk of metastasis and of amputation instead of limb-salvage surgery.
- **England, the surgery.** We know how often a sarcoma is operated on before anyone knows it is a sarcoma. According to PubMed, in 2,201 patients referred to the Royal Orthopaedic Hospital, Birmingham, 402 (18 per cent) had already undergone an unplanned excision elsewhere because the lump was not recognised as a sarcoma; 87 per cent needed a second operation, residual tumour was found in 59 per cent of them, and a positive margin in a high-grade tumour carried a 60 per cent risk of local recurrence even with radiotherapy ([Chandrasekar and colleagues, J Bone Joint Surg Br 2008](#)). So nearly one patient in five has an operation that should not have happened, and more than half of them still have tumour left behind.

The operation that should not happen, and how the model stops it. This is the single most expensive failure in the pathway (Figure 2). A patient has a lump. The surgeon takes it for a benign lipoma and removes it whole. The pathology report then says sarcoma, with tumour at the cut edge. The patient needs a second, wider operation, often radiotherapy, and sometimes an amputation that a planned first operation would have avoided. With the model, the lump is biopsied first, the model reads the H&E and says “sarcoma, refer”, and the first operation is done once, at a specialist centre, with the right margin. One condition must be stated plainly: the model can only help when a biopsy is taken before anything is removed. For a lump that is excised without a biopsy there is no slide to read. The proposal therefore pairs the model with the existing guideline rule, biopsy before excision for any suspicious lump, and measures how often that rule is followed as one of its outcomes.

- **England, the money.** A 2025 cost study from a large English district general hospital put NHS hospital costs at about £11,200 per patient for an early-stage cancer diagnosis and about £23,800 for a late-stage diagnosis ([BMC Cost Effectiveness and Resource Allocation, 2025](#)). Delayed and incorrect cancer diagnoses also cost NHS Resolution £81.5 million in damages and legal fees between April 2021 and March 2023 ([NHS Resolution review, summarised](#)).
- **Taiwan, the rule.** Reimbursement is set by the National Health Insurance Administration under the National Health Insurance Act’s fee schedule. From 1 May 2024 the NHIA pays for next-generation sequencing in 19 cancers at three fixed rates, NT\$10,000 for BRCA, NT\$20,000 for a small panel of up to 100 genes, NT\$30,000 for a large panel, **once per person per cancer type for life**, with an estimated 20,000 patients a year and a budget of about NT\$300 million ([NHIA press release, 2024](#)). Only regional hospitals or above, or hospitals with cancer-care accreditation, listed under the Regulations Governing the Administration of Special Medical Technologies, Examinations, Devices and Equipment as approved laboratory-developed-test sites and running a molecular tumour board may claim it ([NHIA NGS payment Q&A, 2024](#)). District hospitals cannot order it in house.
- **Taiwan, the gap in evidence.** No published figure exists for how many Taiwanese sarcoma patients receive a misrouted first panel, how many referrals are wasted, or how many diagnoses are missed; the scheme is less than two years old and the NHIA has not released utilisation by cancer type. But the rule itself tells us the consequence. If the first panel is the wrong one, the patient’s only reimbursed test is gone, and the second one is paid by the patient or not done at all. Producing those numbers is one of the first deliverables of the Taiwan validation arm.

The bottleneck is not surgical capacity. It is three things that go wrong before the right operation. A lump is removed before anyone knows it is a sarcoma. A patient is referred to the specialist pathway who did not need it. And the surgical plan waits for weeks between the biopsy and the molecular answer. All three start at the same place, the first H&E slide at the referring hospital. The next section shows where a model that reads that slide on

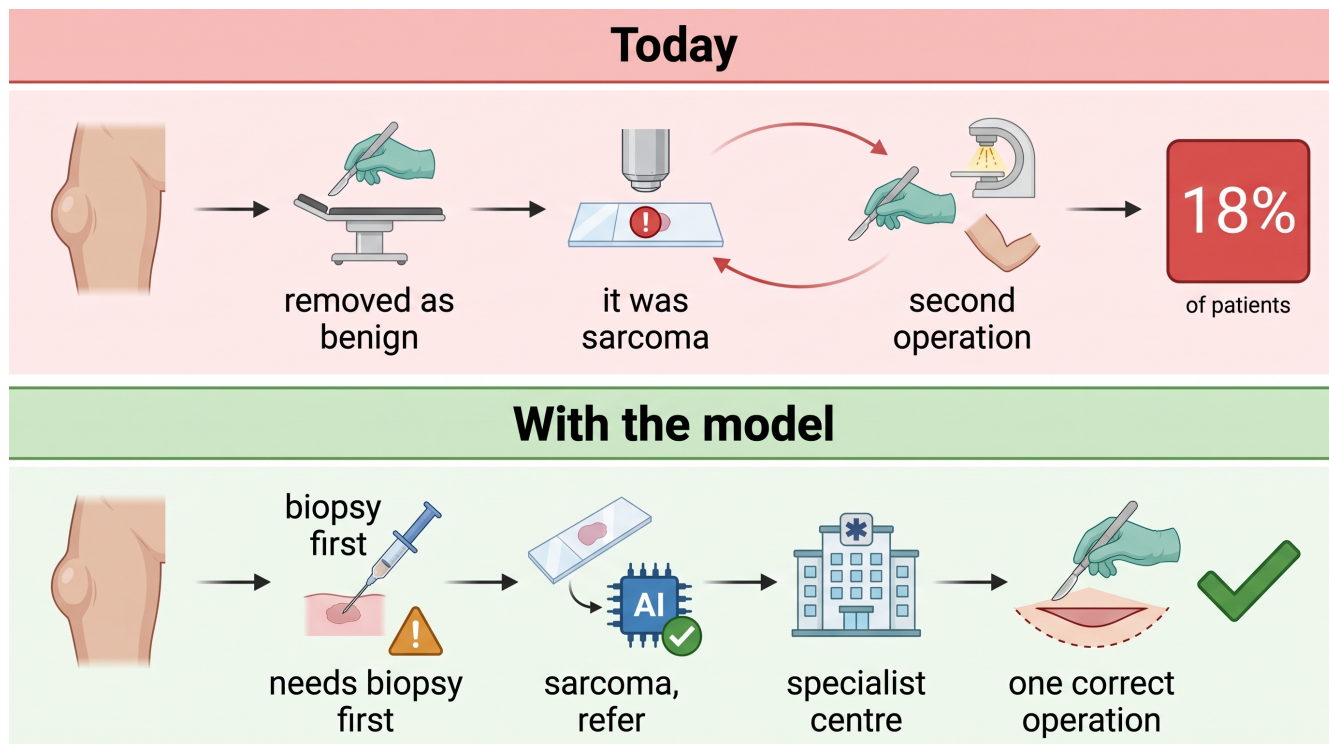


Figure 2: Today: a lump removed as benign turns out to be sarcoma and needs a second operation. With the model: biopsy first, the slide is read, the patient is referred, one correct operation|521x291

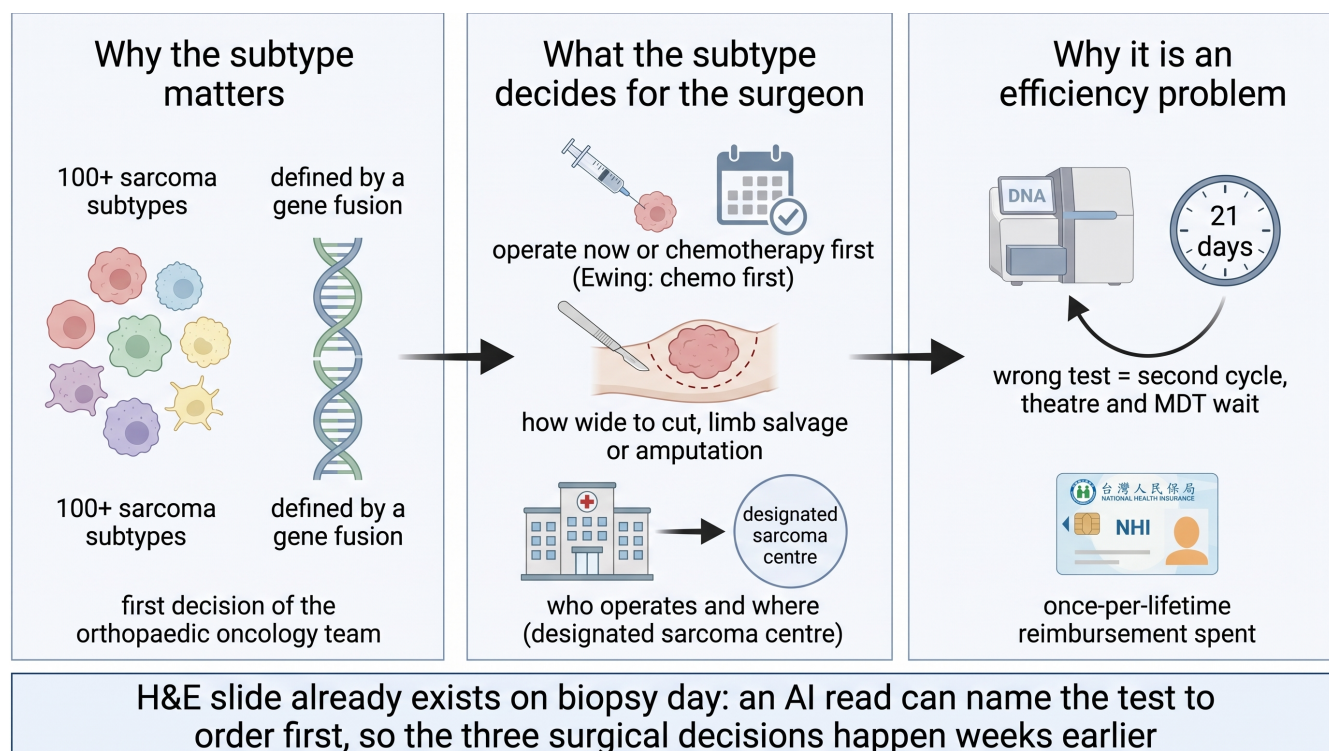


Figure 3: Why the subtype matters, what it decides for the surgeon, and why it is an efficiency problem

the day of the biopsy sits in this pathway.

2. Where the AI sits, and what it returns

Between step 2 and step 3. The H&E slide that already exists is scanned and scored. The output is a probability for each subtype and a recommendation for which genomic test to order first. The pathologist sees the slide before the score. The score never replaces the genomic test and never generates a report on its own.

The model reads the routine H&E slide, inside the viewer the laboratory already uses, and returns within minutes two probabilities: is this a sarcoma, and which subtype.

the goal of it is to On day one it names the panel to order and how urgently, and flags slides that do not match the requested test.

The orthopaedic oncologist therefore settles at the first meeting, not three weeks later, whether to give chemotherapy first, how wide the margin must be, and whether the limb can be saved.

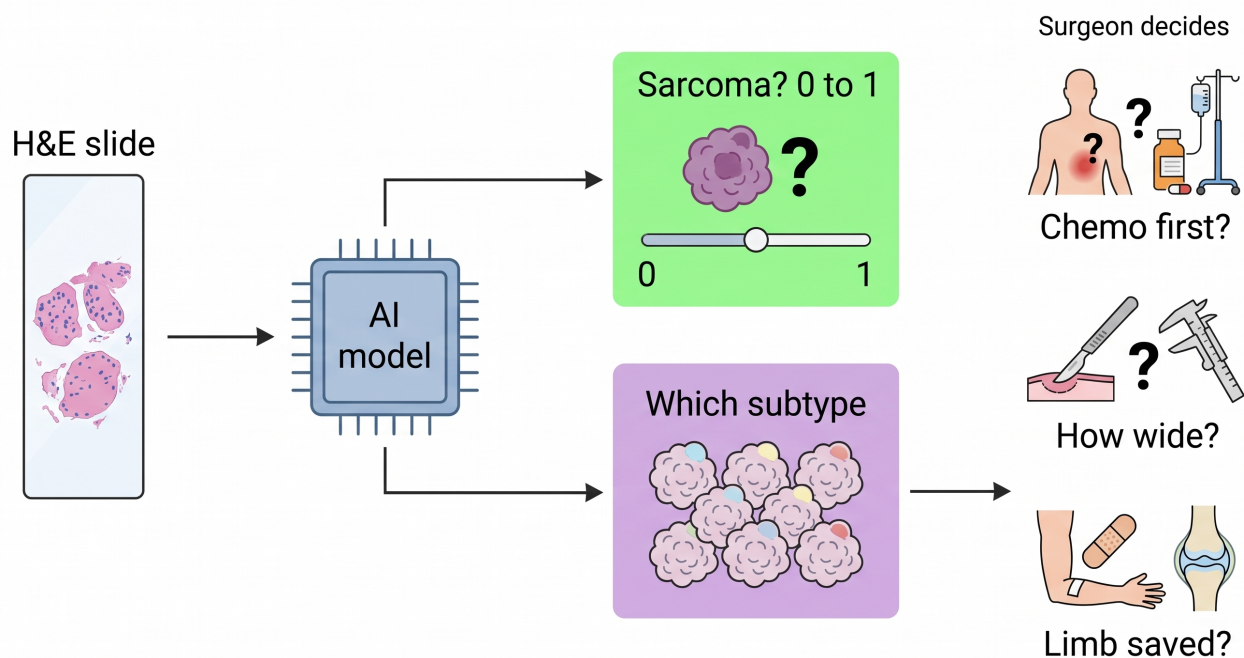


Figure 4: One H&E slide in, two probabilities out, and the three decisions the surgeon can then make

What is in this submission. Figure 4 shows the two outputs. The notebook here does a first step on the second only, two common subtypes on open TCGA data. The production model adds the first output and the fusion-defined subtypes.

Where it sits. Figure 5. Today the slide goes to the pathologist, a test is ordered, and a wrong panel repeats the loop; six weeks is common. With the model, one step is added between slide and pathologist. The pathologist reads the slide first, then the score. One correct test is ordered on day one; three weeks in total.

Efficiency lever: the right genomic test is ordered on day one at the referring hospital, so the sarcoma centre receives a case with the molecular result already in progress.

3. Integration and safety

How it plugs in. No new hardware. The model runs as a module in the existing digital pathology viewer, as third-party AI already does through Sectra Amplifier ([example listing](#)). The result is written into the laboratory system next to the H&E report, with the model version for audit. The slide stays inside the hospital network. A working demo

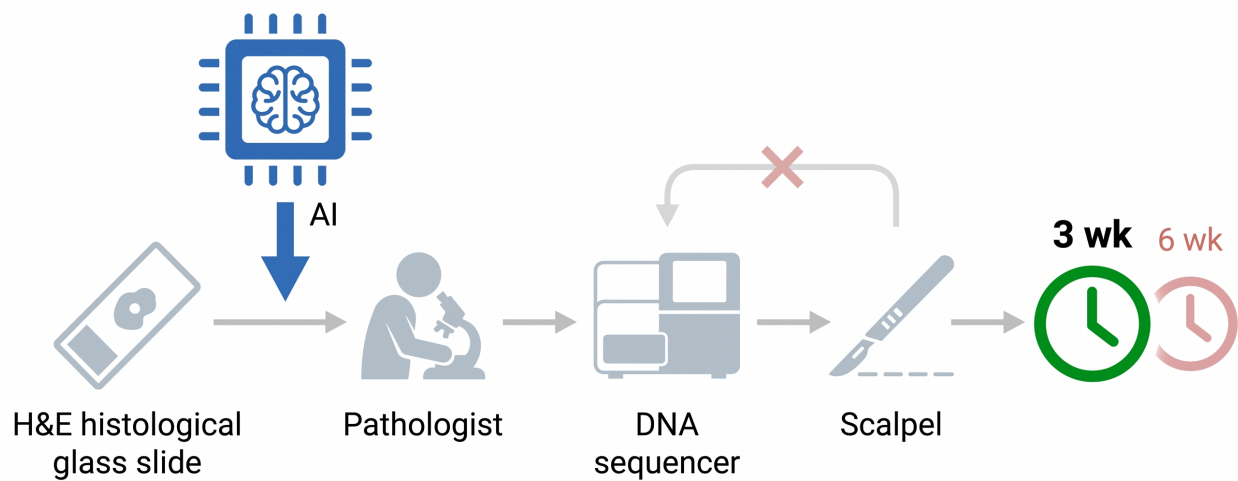


Figure 5: The existing pathway in grey; the AI step inserted in blue; six weeks becomes three

of the baseline, upload a slide and read the result, is live at <https://iressa-sarcoma-triage-demo.static.hf.space/> and in demo/.

The model runs inside the laboratory's existing digital pathology system. Slides do not leave the network. Output is a structured field in the laboratory information system alongside the H&E report, with the model version and confidence recorded for audit. Vendors already provide marketplace integration for third-party AI, for example Sectra Amplifier ([Panakeia on Sectra](#)), so no new viewer is needed.

Figure 6 is a mock-up of that screen. The numbered parts are the only additions to the viewer the pathologist already uses.

Safety design.

- The pathologist reads the H&E first. The score appears after the first read, to limit automation bias.
- Low-confidence cases are labelled "no recommendation" and follow the current pathway unchanged.
- Every score is logged with the model version, the scanner and the staining laboratory, so drift at a single site is visible within weeks.
- The model is retrained only under a documented change-control plan, matching the regulator's expectations for adaptive AI ([TFDA predetermined change control guidance](#); the MHRA takes the same approach through the AI Airlock, [GOV.UK](#)).

4. Taiwan

The same tool at a Taiwan medical centre. National Health Insurance reimburses next-generation sequencing once per lifetime, capped at NT\$30,000, and only regional hospitals and above may perform it ([Health Promotion Administration](#)). District hospitals therefore refer without a molecular result. An H&E triage score tells them whether and where to refer, and protects the single reimbursed test from being spent on the wrong panel.

5. Diversity and equity

Sarcoma AI has been trained almost entirely on European and North American slides. Performance is reported separately by laboratory, scanner, sex, age group and ancestry, and the Taiwan cohort exists so that an Asian

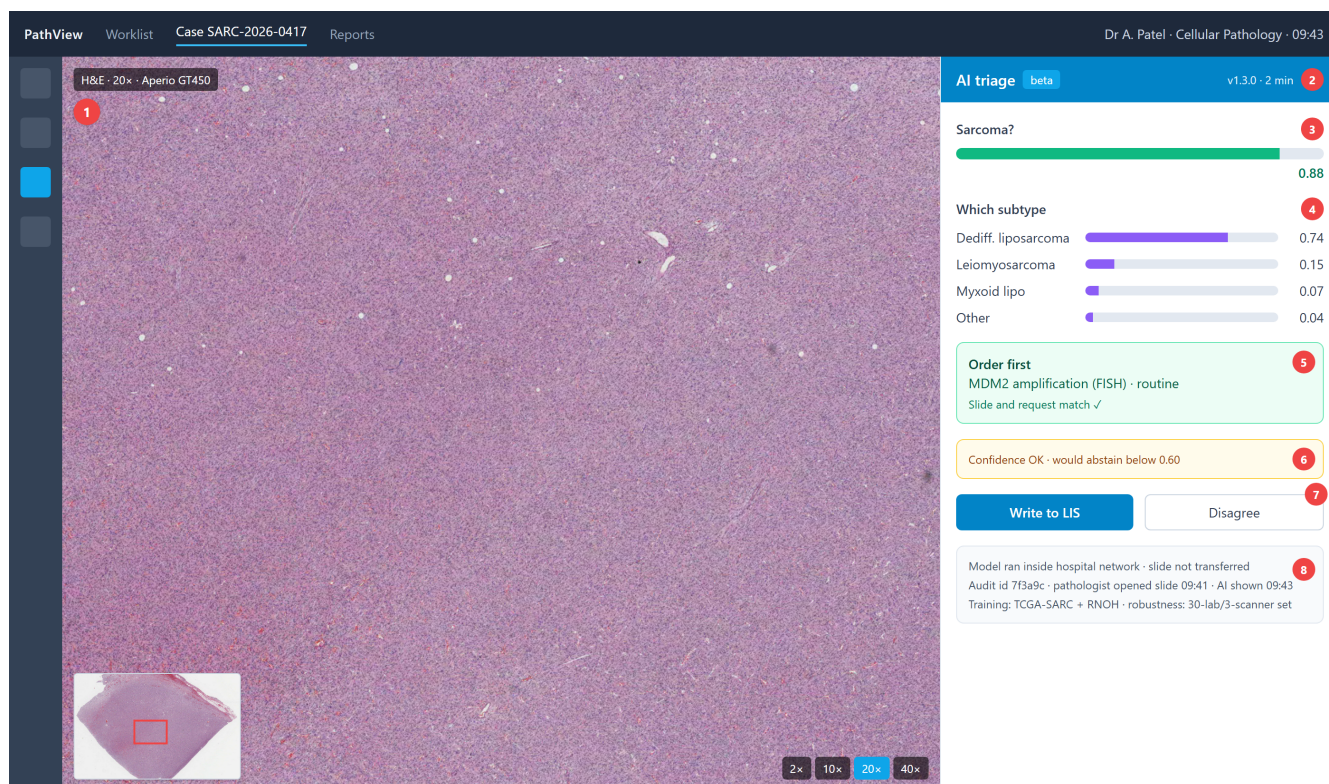


Figure 6: What the pathologist sees: (1) the slide, (2) the AI panel with model version, (3) sarcoma probability, (4) subtype probabilities, (5) the panel to order first and whether slide and request match, (6) the abstain threshold, (7) write to the laboratory system or disagree, (8) audit trail showing the slide never left the network and that the pathologist read first

population is in the validation set from the start, not as an afterthought.

Patient-level data are never moved between countries or institutions. Only the model moves.

6. Clinical implementation, stage by stage

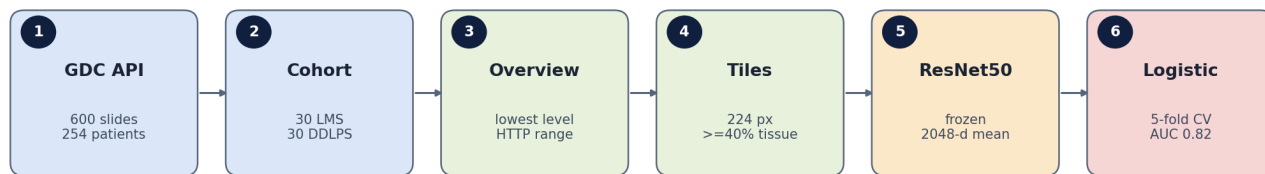
Stage	What happens	Who owns it	Measure of success
Retrospective validation	Model run on archived cases with known fusion status at one sarcoma centre and on the 30-hospital stain-variation dataset	Research team with the centre's pathology department	Per-subtype AUC, per-laboratory AUC, abstention rate
Shadow mode	Model scores live cases, output recorded, not shown to clinicians	One Genomic Laboratory Hub	Misroute rate with and without the model, measured on the same cases
Advisory mode	Score shown to the specialist pathologist as a second opinion at the point of test ordering	Sarcoma centre pathology	Days from biopsy to molecular result, repeat-test rate
Referral triage	Score available to the referring general pathologist	Regional pathology network	Proportion of referrals arriving with the correct test already ordered

Technical part: how the baseline model was built

The notebook `01_sarcoma_hne_triage.ipynb` section by section: data, tiling, features, training, errors, the browser demo, limitations, scope, and how to run it.

1. Pipeline at a glance

Everything in this section is in one notebook, `technical/01_sarcoma_hne_triage.ipynb`, which runs end to end on a laptop CPU in about an hour, most of it download time. Figure 7 shows the six steps. Each step is one notebook section, with the story above the code, a comment on every line, and the output explained below it.



One slide per patient, seed 42, CPU only. Steps 1-6 = notebook sections 2-7.

Figure 7: The six steps of the baseline, one notebook section each

The design choice behind every step is the same: the smallest thing that proves the overview image carries subtype signal, so that a reviewer can rerun it and get the same number. Nothing is trained from scratch, nothing needs a GPU, and the random seed is fixed at 42.

2. Data, open TCGA-SARC slides

Source. The Cancer Genome Atlas sarcoma project (TCGA-SARC) holds diagnostic whole-slide images that are open access through the Genomic Data Commons API ([GDC TCGA-SARC](#)). Notebook section 2 lists every diagnostic slide with its file size, patient and primary diagnosis.

Listed	Count
Diagnostic slides	600
Patients	254
Leiomyosarcoma (LMS) slides	142
Dedifferentiated liposarcoma (DDLPS) slides	71

Why these two subtypes. They are the two largest well-defined groups in TCGA-SARC, and they lead to different first tests in practice: DDLPS is confirmed by MDM2 amplification, LMS by smooth-muscle immunohistochemistry. The fusion-defined subtypes that are the clinical goal of the project (Ewing, synovial, myxoid liposarcoma) have too few open slides to train on today.

Cohort. One slide per patient, so no patient can appear in both the training and the test fold. Thirty patients per class. We took the thirty smallest files per class because the lowest pyramid level scales with file size and reads faster. This is a selection bias and it is stated as such in section 8.

What is read. Only the lowest pyramid level of each slide, fetched over HTTP range requests with `fetch_overviews.py`, about 30 seconds per slide. The whole dataset is sixty small PNG files instead of sixty gigabytes, which is why anyone can rerun the notebook. Figure 8 shows two overviews per class.

3. Tiling and tissue mask

ResNet50 expects 224 px squares, so notebook section 5 slides a 224 px window with no overlap across each overview. A tile is kept only if at least 40 per cent of its pixels are darker than near-white (grey value below 220), which removes glass and most pen marks. Slides with fewer than three tissue tiles would be dropped; none were.

Class	Slides	Mean tiles	Min	Max
DDLPS	30	70.5	3	129
LMS	30	48.1	6	105

DDLPS slides give more tiles because the tumours are larger and fattier. The count itself cannot leak into the classifier, because the next step averages all tiles into one vector per slide (Figure 9).

4. Features, frozen ResNet50

Notebook section 6 passes every tile through a ResNet50 with ImageNet weights (torchvision IMAGENET1K_V2), with the classification head removed, so each tile becomes a 2048-dimensional vector. The vectors of one slide are averaged, giving a 60 by 2048 matrix, one row per patient. The CNN is not trained or fine-tuned at any point.

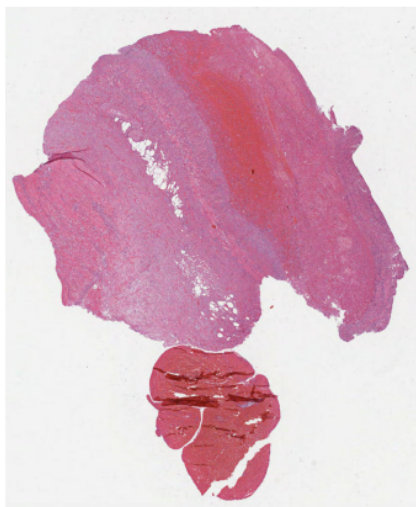
This is the cheapest credible baseline. It runs on a CPU in minutes and has no hyperparameters to tune, so the result cannot be an artefact of tuning on sixty patients. The production model replaces this step with a pathology foundation model on full-resolution tiles (section 8).

5. Training and cross-validation

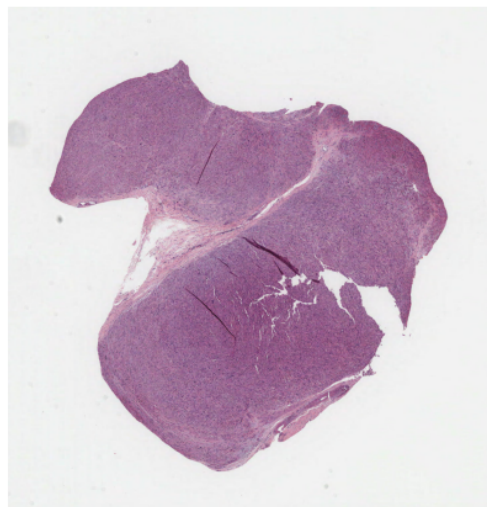
Sixty patients cannot support anything bigger than a linear model. Notebook section 7 fits a standard scaler and a logistic regression ($C = 0.1$, L2 penalty) inside a 5-fold stratified cross-validation, seed 42. Stratification keeps the class ratio in every fold, and one slide per patient means no patient is in both train and test. Every number reported is out-of-fold: the model that scored a patient never saw that patient.

Slide overviews (lowest pyramid level)

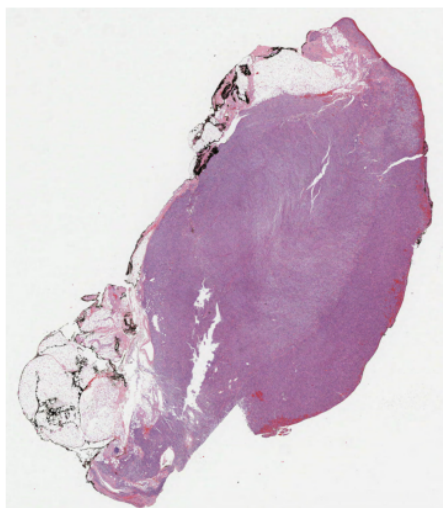
TCGA-FX-A48G LMS



TCGA-3B-A9HY LMS



TCGA-3R-A8YX DDLPS



TCGA-FX-A3NK DDLPS

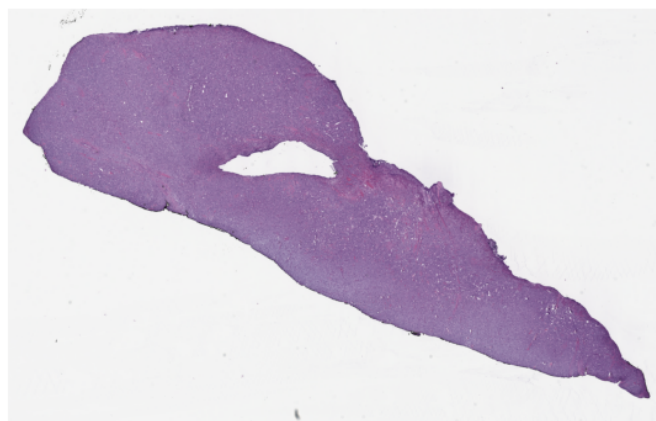


Figure 8: Slide overviews, lowest pyramid level, two per class

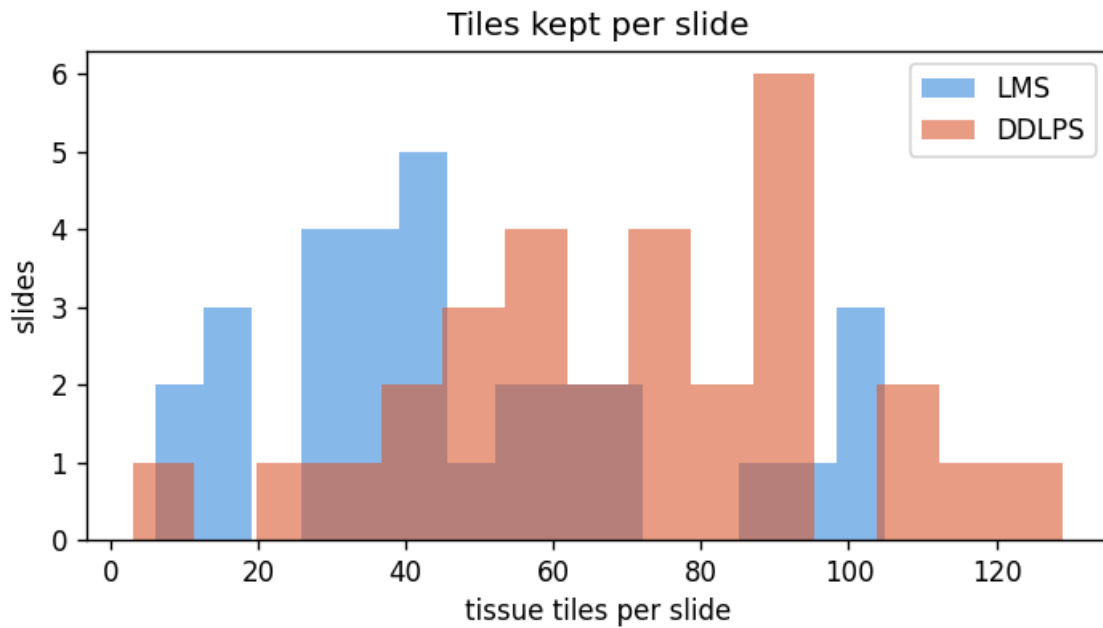


Figure 9: Tissue tiles per slide by class

Fold	AUC
0	0.778
1	0.722
2	1.000
3	0.917
4	0.611
Out-of-fold, all 60	0.822

At a 0.5 threshold 42 of 60 patients are correct, nine errors in each class (Figure 10). The spread between folds, 0.61 to 1.00, is what twelve-patient test folds look like. It is a limit of the cohort size, not a bug.

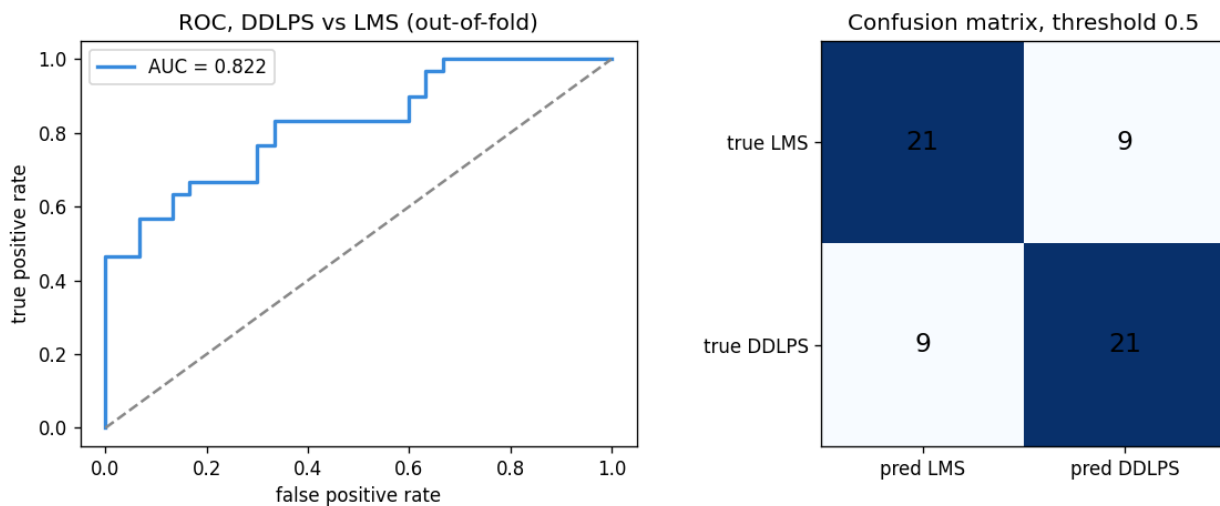


Figure 10: ROC curve and confusion matrix, out-of-fold

6. Where it fails

Notebook section 8 ranks the patients by how confidently wrong the model was. All four worst cases are DDLPS slides scored as LMS with $p(\text{DDLPS})$ below 0.08 (Figure 11).

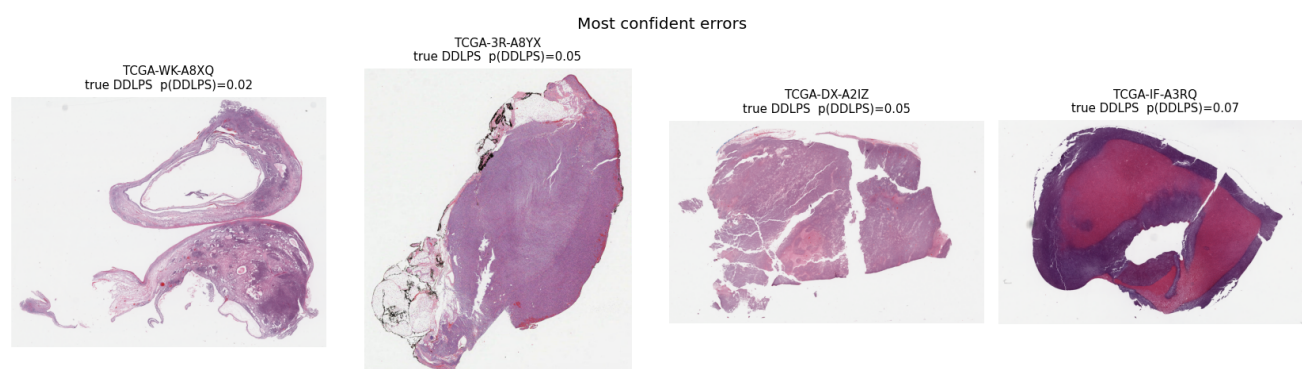


Figure 11: Most confident errors, all DDLPS scored as LMS

onnxruntime-web: tile at 224 px, keep tiles with at least 40 per cent tissue, extract features, average, apply the logistic head. The image never leaves the machine it is opened on. Quantisation shifts probabilities by a few hundredths compared with the notebook.

The demo shows what the surgeon or pathologist sees: the two subtype probabilities, the first test the model would order, and a low-confidence flag when the top probability is below 0.60, in which case the case follows the standard pathway.

8. Limitations and the production model

Limitations, stated plainly.

1. Sixty patients and two subtypes. This proves the pipeline runs end to end on open data. It is not a clinical result.
2. Overview resolution only, roughly 16 times downsampled from the scan. Nuclear detail is invisible.
3. Smallest-file selection bias. The sixty slides are the smallest per class, which may favour smaller or less complex tumours.
4. A single scanning programme (TCGA). Stain and scanner variation between hospitals is untested.
5. Mean pooling. Informative regions are averaged with uninformative ones, which is exactly what section 6 shows.

What changes in the production model. The table maps each limitation to the change that removes it.

Baseline (this notebook)	Production model
Overview, lowest pyramid level	Full-resolution tiles at 20x
Frozen ImageNet ResNet50	Pathology foundation model (UNI or TITAN)
Mean pooling	Attention pooling, multiple-instance learning
60 TCGA patients, 2 subtypes	Multi-hospital cohort, fusion-defined subtypes included
Cross-validation only	External validation on held-out hospitals, calibration, subgroup audit

The multi-hospital validation set and the robustness gate before any clinical use are described in the business case.

Reporting standard. The baseline is reported against the items of TRIPOD+AI, the 2024 reporting guideline for prediction models that use machine learning ([Collins and colleagues, BMJ 2024](#)): data source and eligibility (section 2), predictors and outcome (sections 3 to 5), sample size and its limits (section 8), model building and internal validation (section 5), performance with calibration not yet assessed, and the fairness items not yet assessable on sixty TCGA patients. A completed TRIPOD+AI checklist will accompany the production model, not this prototype.

9. Scope of the prototype, and what the production model adds

The code in `technical/` is a proof of concept on open TCGA-SARC data, built so that any reviewer can rerun it on a laptop and get the same numbers. It shows that the H&E image alone separates two common sarcoma subtypes. The production model is a larger version of the same pipeline, not a different product. Section 8 sets the two side by side.

Moving from the prototype to the production model changes the data and the model. It does not change where the tool sits in the pathway or who acts on its output.

10. How to run

```
conda create -n sarc python=3.12
pip install -r technical/requirements.txt
cd technical
jupyter lab 01_sarcoma_hne_triage.ipynb # run all, first run downloads ~60 overviews
jupyter lab 02_savings_model.ipynb
```

To see the trained baseline run on a slide, open the live demo at <https://iressa-sarcoma-triage-demo.static.hf.space/> (hosted as a Hugging Face Space; the model runs in your browser and the image never leaves your machine), or run it locally:

```
cd demo
python -m http.server 8765
```

Then open <http://localhost:8765/>, drop an H&E overview image or click one of the six bundled TCGA-SARC slides. The model runs in the browser in a few seconds and shows the subtype probabilities and the panel it would order first. `demo/README.md` explains the files.

Seed 42 is fixed in both notebooks. Features and overviews are cached under `technical/data/` after the first run.

Business part: business case

Structured on the five-case model that NHS and HM Treasury business cases use ([HM Treasury Green Book, five-case model](#)): strategic, economic, commercial, financial, management. Sections are numbered in reading order; the map in section 1 shows the five parts on one page.

Part A. Strategic case: the problem, the solution, and why now

The case for change. Sarcoma subtype decides the operation and the oncology plan, the confirming genomic test is centralised and slow, and a wrong first order costs a full test cycle in England and the single reimbursed test in Taiwan. We propose decision support that reads the H&E slide already in hand and makes the first genomic order the right one. This part states the problem, the solution, and what the submitted prototype does and does not prove.

1. Map of this business case

The five parts follow the five-case model used for NHS and HM Treasury business cases ([Green Book](#)). Each part answers one question, and the sections under it are read in order (Figure 12).

Part	Question it answers	Sections
A. Strategic	What is wrong today and what we propose; the clinical evidence is in the Medical part	1 to 3
B. Economic	What the payer saves and what one slide costs to run	4, 5
C. Commercial	Who pays, how the money flows, how big the market is, who else sells	6 to 9

Part	Question it answers	Sections
D. Financial	What three years cost and who funds them	10 to 12
E. Management	How it is rolled out, by when, under which regulation, with patients involved how, by whom, and what could go wrong	13 to 18

Key performance indicators: misroute rate, days to molecular result, abstention rate and per-site AUC, listed in notebook 02 section 5 and notebook 01 section 9.

2. The problem we are solving

The bottleneck is not surgical capacity. It is three things that go wrong before the right operation. A lump is removed before anyone knows it is a sarcoma. A patient is referred to the specialist pathway who did not need it. And the surgical plan waits for weeks between the biopsy and the molecular answer. All three start at the same place, the first H&E slide at the referring hospital. The clinical evidence, the pathway and the costs of each failure are set out in the Medical part, section 1.

3. The solution, as a product

A decision-support module inside the digital pathology viewer the laboratory already uses. It reads the routine H&E slide and returns two probabilities, is this a sarcoma and which subtype, with the genomic panel to order first. Where it sits in the pathway, what the pathologist sees, and how it is integrated and kept safe are in the Medical part, sections 2 and 3.

How it is built, and why the data meet the diversity and ethics criteria. Training: open TCGA-SARC slides, then Royal National Orthopaedic Hospital slides under the biobank's existing ethics approval. Robustness: a UK dataset in which one tissue block was stained in 30 hospitals and scanned on 3 scanners. External validation: Taiwanese slides from the National Biobank Consortium, the first Asian cohort in any sarcoma AI. Results are reported by laboratory, scanner, sex, age and population. Patient data never crosses a border; only the model moves.

Aim. The first genomic request is the right one for every sarcoma patient. Conservatively, about 2,900 patient-days of waiting removed a year in England and about 20 once-only reimbursed tests protected a year in Taiwan (`technical/02_savings_model.ipynb`). Payer and pricing are in sections 4, 5 and 6.

Efficiency lever: **one H&E slide plus one model call replaces up to three weeks of waiting and one avoidable round of testing per misrouted case.**

Part B. Economic case: what it saves and what it costs to run

Value for money. The direct laboratory saving is modest because the test is cheap; the value is in patient-days of waiting removed and operations planned with the subtype known. Inference cost per slide is negligible, so the cost of the service is validation, regulation and support. Every input is labelled sourced or assumption and can be changed in the savings notebook.

4. How much money it saves, and why a government payer pays

The arithmetic is in `technical/02_savings_model.ipynb`, with every input labelled SOURCED or ASSUMPTION so a reviewer can change any number and re-run.

Input	Value	Status
New sarcoma cases, UK, per year	about 5,900 (Sarcoma UK)	sourced
Cost of one NHS NGS cancer panel	£339 (2017 figure , historic)	sourced, dated
Urgent solid tumour panel target	90 per cent within 21 days (East Genomics)	sourced

Input	Value	Status
Taiwan NHI NGS reimbursement	once per lifetime, cap NT\$30,000 (HPA summary)	sourced
Share of cases needing a molecular test	0.50	assumption
Share of those tests misrouted or repeated	0.10	assumption
Share of misroutes the triage catches	0.70	assumption
Days saved per caught misroute	14	assumption

With those conservative assumptions the UK direct laboratory saving is about £70,000 a year and about 2,900 patient-days of waiting are removed. The direct saving is small because the test is cheap. The value is in the days, and in Taiwan in protecting a test that can only be reimbursed once.

Why the payer is the government and not the hospital:

1. In England cancer genomic tests are funded centrally by NHS England, so the avoided repeat and the triage licence sit on the same budget ([BWC genomics](#)).
2. The 21-day target is a national performance measure for the Genomic Laboratory Hubs, so a tool that removes second cycles is bought to protect a target.
3. Taiwan NHI pays once per lifetime, so a wrong first order is an unrecoverable public cost.
4. Sarcoma is rare and referral is centralised, so one national licence covers the whole pathway.

The two assumptions a pilot must measure first are the misroute rate and the catch rate. The sensitivity plot in the notebook shows every input moves the result proportionally.

5. Unit economics

Per-slide inference cost, computed from public prices. Feature extraction with a pathology foundation model such as UNI takes about 2 to 5 minutes per whole slide on one GPU ([Scientific Reports 2025 retrieval study](#), [ensemble study](#), [PMC](#)). An AWS g5.xlarge with one A10G costs about US\$1.01 an hour on demand ([g5.xlarge pricing](#)).

Item	Value	Basis
GPU time per slide	5 minutes, upper bound	UNI extraction time above
GPU cost per slide	about US\$0.08	$5/60 \times \$1.01$
Storage per slide	1 to 8 GB, about 2 GB compressed	MDPI review of WSI
Storage cost	a 250,000-slide-a-year laboratory spends about US\$90,000 a year	Digital Pathology Association
Volume, UK	about 3,000 slides a year if half of new cases are triaged	\$4

At that volume compute is under US\$300 a year. The cost of the product is regulatory, clinical validation and support, not inference. A per-slide price in the tens of pounds is therefore almost pure margin once the device is certified, and a national licence should be priced against the days saved, not against compute.

Part C. Commercial case: who buys, how they pay, and who else sells

The buyer and the deal. One payer in each country, NHS England through the Genomic Laboratory Hubs and the National Health Insurance Administration in Taiwan, and two kinds of user who pay nothing extra. The product reaches them inside the digital pathology viewer they already use. No company sells an H&E-to-fusion product for sarcoma today.

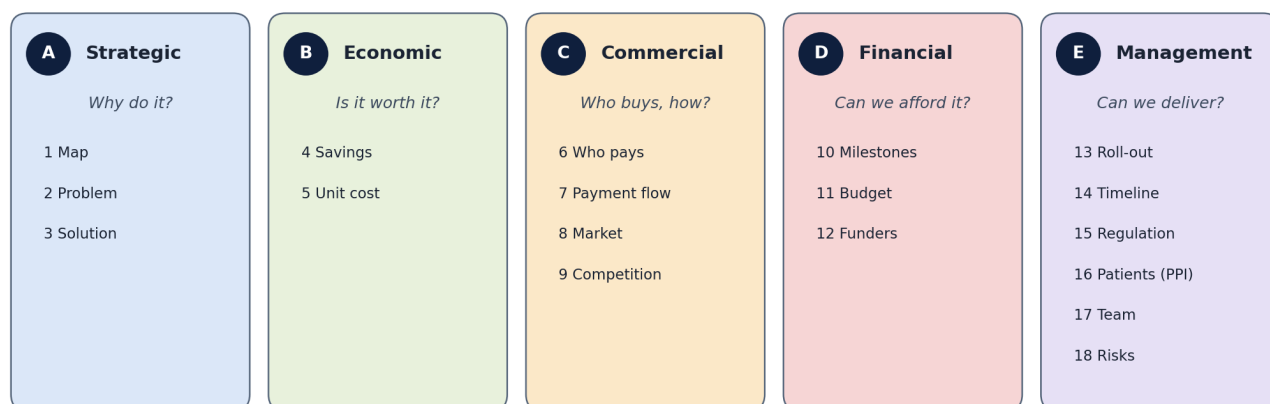


Figure 12: The five parts and the sections under each

6. Who pays, and why they would

One payer in each country, two kinds of user, and one channel. Figure 13 shows the money and the use separately.

Who	Role	What they buy	Why it is worth it to them
NHS England, through the Genomic Laboratory Hubs	Primary payer, England	Per-slide licence, or an annual regional subscription, delivered inside the hub's existing digital pathology feed	The hub pays for every genomic test from a central budget. Each misrouted request costs a repeat panel and a second three-week cycle against a 21-day national target. One licence replaces both.
Taiwan National Health Insurance Administration	Primary payer, Taiwan	A reimbursement code for AI-assisted pathology triage	NGS is paid once per patient per lifetime, capped at NT\$30,000. A wrong first panel spends it. Triage protects it.
Specialist sarcoma centres (Royal National Orthopaedic Hospital, Birmingham, Oxford, Newcastle)	User, and the clinicians who ask the hub to buy	Nothing extra; the result arrives with the referral	The orthopaedic team has the subtype before the first multidisciplinary meeting, so the theatre booking, the margin and the limb-salvage decision are made three weeks earlier.
District and regional hospitals (both countries)	User	Nothing extra; the score appears in the viewer they already use	They cannot run genomic tests themselves. The score tells them whether to refer, and stops the lump being removed before anyone knows it is a sarcoma.

Who	Role	What they buy	Why it is worth it to them
Digital pathology marketplaces (Sectra Amplifier, Leica)	Channel	A listed module, paid by revenue share	Scanner vendors do not buy rare-cancer models outright. They list them and take a share, which is how third-party pathology AI already reaches hospitals (Sectra Amplifier listing).

Price against cost. Compute is under £0.10 a slide (section 6). One repeat genomic panel is about £339 ([2017 NHS figure](#)). One unplanned excision followed by re-operation costs the system far more than that; management after an unplanned excision cost 64 per cent more than after a planned one in a Scandinavian series ([EJSO 2020](#)). A per-slide fee in the tens of pounds therefore pays for itself on the first avoided repeat test, and many times over on the first avoided re-operation. The fee used in the financial projections is £40 a slide; it is an assumption until a hub quotes.

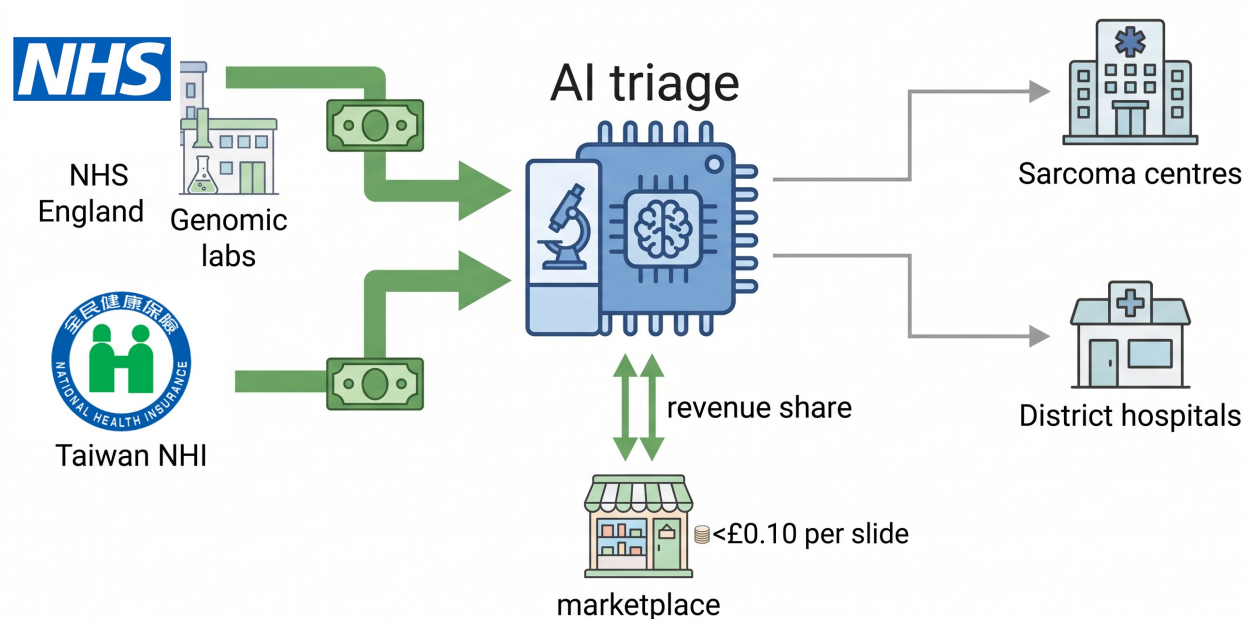


Figure 13: Business model: who pays (green, money in), who uses (grey, result out), and the marketplace channel

7. Payment pathway

Figure 14 shows who pays whom along the diagnostic pathway. Grey arrows carry the sample and the report, green arrows carry money. The source file is `business/payment_pathway_biorender.jpg` ([editable BioRender version](#)).

Flow: patient biopsy at a local hospital → H&E slide scanned → model call (charged per slide to the laboratory or hub) → triage report → correct genomic test ordered once → result reaches the sarcoma multidisciplinary team.

8. Market size

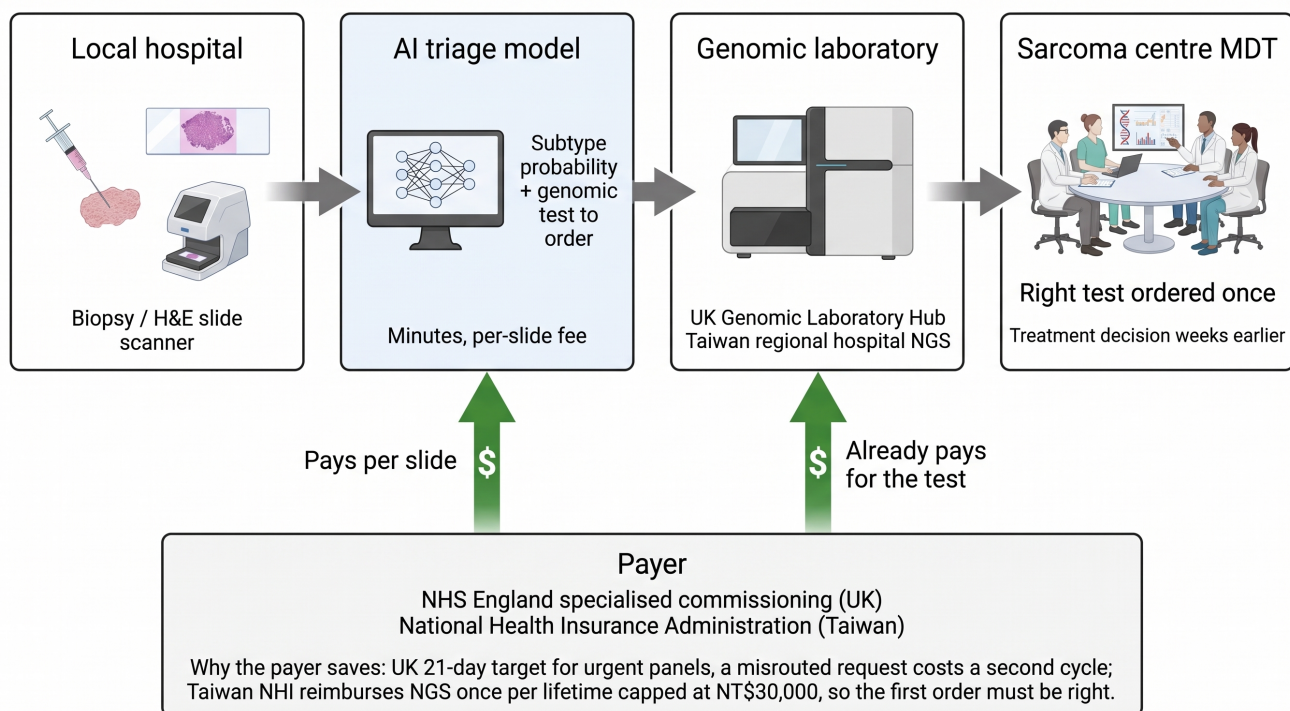


Figure 14: Payment pathway: sample and report in grey, money in green

Market	Cases per year	Source
UK, all sarcoma	about 5,900 (700 bone, 5,200 soft tissue)	Sarcoma UK statistics
Taiwan, primary bone cancer	1,238 cases in 2003 to 2010, about 155 a year, age-standardised rate 6.70 per million	Hung and colleagues, Ann Surg Oncol 2014
Taiwan, soft tissue sarcoma	3,843 cases in 2003 to 2011, about 430 a year, age-standardised rate 1.63 per 100,000	Medicine (Baltimore) 2015, Taiwan Cancer Registry

The Taiwan figures are a decade old and are used as a floor. The current annual report of the Taiwan Cancer Registry is the source to update from ([Health Promotion Administration registry reports](#)). Laboratories: seven Genomic Laboratory Hubs in England ([BWC genomics](#)); Taiwan NGS is restricted to regional hospitals and above ([HPA summary](#)).

9. Competition

Nobody markets an H&E-to-fusion-gene product for sarcoma today. The category exists in other cancers, which proves the regulatory route and the buyer, and leaves sarcoma open.

Company	Product	What it predicts from H&E	Regulatory status	Relevance
Owkin	MSIntuit CRC	Microsatellite instability in colorectal cancer, rules out about half of MSS patients at 96 per cent sensitivity	CE-marked (EurekAlert , Nature Communications validation)	Closest business model, a pre-screen that reduces molecular tests
Panakeia	PANProfiler Breast	ER, PR, HER2 status	UKCA and CE (Panakeia)	UK company, same regulatory path we plan
Paige	Prostate Biomarker Suite, Paige Predict	AR amplification, TP53, RB1, PTEN in prostate; 123 biomarkers across 16 tumour types	CE-IVD and UKCA for prostate (Business Wire 2022); Predict launched under Tempus (Stock Titan)	Largest player, no sarcoma model listed
Academic, UCL and RNOH	AI Scope sarcoma diagnosis, UKRI funded	Subtype from pathology plus genomics	Research (UCL news 2023)	Partner, not competitor

Academic literature on sarcoma AI is prognosis-first, not fusion-first, for example survival prediction from H&E with reported AUC 0.97 in one cohort ([PMC review](#)) and margin-aware prognosis in 2025 ([Scientific Reports 2025](#)). A 2025 review frames fusion detection as the next AI target in soft tissue sarcoma genomics ([PubMed 41497157](#)).

Part D. Financial case: budget and funders

Affordability. Three years from prototype to first licence, grant-funded so that the first contract is signed with evidence rather than a price guess. The budget is itemised line by line and each line is marked sourced or assumption. The first application is the Cancer Research UK Early Detection and Diagnosis Primer Award.

10. Milestones and funding

Year	Milestone	Funding route	Amount and source
1, 2026 to 2027	Production model on RNOH and TCGA slides, robustness on the 30-hospital set, first manuscript	NIHR i4i Product Development Award	No upper limit, typical £0.5 to £1.5 million (NIHR i4i PDA); next round October 2026 (RedKnight note)
1, parallel	Feasibility and business case	SBRI Healthcare phase 1	Up to £100,000 for six months (SBRI Healthcare cancer programme)
2, 2027 to 2028	Shadow deployment at one hub, DTAC, NICE ESF evidence	Innovate UK Biomedical Catalyst	£25 million pool for industry-led small projects (Innovate UK)
2, parallel	Taiwan validation cohort	NSTC smart-healthcare innovation programme 2024 to 2027 (NSTC)	Submission under the TFDA AI/ML software guidance (TFDA guidance)
3, 2028 to 2029	UKCA technical file, first paid licence	Licence revenue, SBRI phase 2	

11. How much money, line by line

Three-year budget. Sourced lines carry a link; the rest are labelled assumptions and should be replaced by quotes before any grant is submitted.

Line	Year 1	Year 2	Year 3	Basis
Machine learning researcher, 1.0 FTE	£60,000	£62,000	£64,000	Oxford research grade 7, £39,424 to £47,779 salary (Oxford salary scales), plus about 30 per cent on-costs, assumption
Clinical lead time, 0.2 FTE	£15,000	£15,000	£15,000	Assumption, buy-out of clinical sessions
Pathology annotation and slide scanning	£20,000	£10,000	£5,000	Assumption, scanning and pathologist time at RNOH
GPU compute	£2,000	£3,000	£3,000	About US\$0.08 per slide at g5.xlarge rates (AWS g5.xlarge); mostly training, not inference
Taiwan cohort, specimen processing fees	£0	£15,000	£5,000	National Biobank Consortium publishes a fee schedule (NBCT application area); amount is an assumption until quoted
Regulatory consultancy and Approved Body audit	£0	£20,000	£40,000	Third-party UKCA assessment is charged at £200 to £500 an hour and totals from the low thousands upward for a Class IIa software device (UKCA guide); MHRA registration about £300 a year per device category from April 2026 (Casus Consulting); clinical safety officer time is the larger part, assumption
Health economics and NICE ESF evidence	£0	£15,000	£10,000	Assumption, contracted

Line	Year 1	Year 2	Year 3	Basis
Patient and public involvement	£3,000	£3,000	£2,000	Assumption, sarcoma charity partnership
Travel, dissemination, open-access fees	£5,000	£7,000	£7,000	Assumption
Total	£105,000	£150,000	£151,000	About £406,000 over three years

The number to defend is the regulatory line. Everything else is ordinary research cost. If the Approved Body quote comes back higher, the roll-out shifts by a quarter, not the model.

12. Where the money comes from, and who is actively giving

Ranked by fit, with the next deadline where it is published.

Funder and scheme	Amount	Fit	Deadline or status	Source
Sarcoma UK, Open Grant Round 2026	Pilot up to £75,000, Project up to £200,000	High, sarcoma-specific, diagnosis is a named priority	Round closed 10 September 2026, next round 2027	Sarcoma UK open grant round
Sarcoma UK, Improving Sarcoma Diagnosis fund	Per call	Highest, the call names this problem	Rolling, see the Sarcoma UK site	Improving sarcoma diagnosis
Sarcoma UK, Early Career Sarcoma Research Fund	Per call	High, DPhil students eligible	Rolling, see the Sarcoma UK site	Early career fund
Bone Cancer Research Trust, Idea Grant and Ewing sarcoma project call	Project call up to £250,000	High for the Ewing arm, a fusion-defined subtype	2026 calls closed 31 July and 1 June; annual	BCRT for researchers
Cancer Research UK, Early Detection and Diagnosis Primer Award	Up to £100,000, one year	High, pays for exactly the year-1 feasibility work	Rolling rounds	CRUK primer award
NIHR i4i Product Development Award	No upper limit, typically £0.5 to £1.5 million	High for years 2 to 3, covers regulatory and clinical evaluation	Round opens October 2026	NIHR i4i PDA
SBRI Healthcare, NHS Cancer Programme	Phase 1 up to £100,000 for six months	Medium to high, NHS-facing feasibility	Open calls announced on the site	SBRI Healthcare cancer programme
Innovate UK Biomedical Catalyst	£25 million pool, industry-led small projects	Medium, needs an SME or spin-out partner	Annual	Innovate UK

Funder and scheme	Amount	Fit	Deadline or status	Source
EMERGE RehabTech Pacesetter (seen at ICNR 2026)	Phase 3 awards up to £30,000 for market analysis, grants 1 January to 30 June 2027	Low to medium, rehabilitation technology focus, but the training and mentoring phases are open to UK clinicians and the business-plan coaching is directly useful	Round 1 closed 19 June 2026, Phase 1 workshop 14 July 2026, next round not yet announced; contact EMERGErehabtech@ntu.ac.uk	Medilink Midlands, EMERGE
Taiwan NSTC smart-healthcare innovation programme 2024 to 2027	Varies; bilateral AI calls about NT\$5 million a year per project	High for the Taiwan validation arm	Calls listed on the NSTC site	NSTC
MHRA AI Airlock, phase 3	Not a grant, a regulatory sandbox	High for de-risking the device route	From April 2026, three years	GOV.UK AI Airlock

Funding plan in one line. We will start by applying for the Cancer Research UK Early Detection and Diagnosis Primer Award, up to £100,000 for one year, because it is the fastest route to a first grant, it carries the lowest entry barrier for an early-career applicant, and it covers the year-1 budget almost exactly. Sarcoma UK's pilot grant (£75,000) is the parallel application in the next round, and SBRI phase 1 tops up year 1. Years 2 and 3 are covered by one NIHR i4i award. Taiwan is covered by NSTC. The three-year total of about £406,000 fits inside that stack with margin.

Who is actively giving right now. Sarcoma UK and the Bone Cancer Research Trust run annual sarcoma-specific calls and have named diagnosis as a priority. Cancer Research UK's primer award is the fastest route to a first £100,000. The EMERGE Pacesetter seen at ICNR is a business-plan accelerator rather than a research funder, and it is rehabilitation-focused, so it is a place to learn the pitch, not the place to fund the model.

Part E. Management case: delivery, regulation, team and risks

Deliverability. Research use, then shadow deployment at one hub, then regulated use as UKCA-marked Class IIa decision support that a pathologist confirms. Performance is reported by laboratory, scanner and population, and the single largest risk, a model that learns stain colour instead of biology, has a measured gate before any clinical use.

13. Adoption and roll-out

1. **Year 1, research use.** Validate on the Royal National Orthopaedic Hospital biobank and on the 30-hospital stain-variation dataset released by the UCL sarcoma group, which shows whether the model survives the change of laboratory and scanner ([Chai, Chen and colleagues, J Pathol Clin Res 2026](#)).
2. **Year 2, shadow deployment** at one Genomic Laboratory Hub, model output recorded but not acted on, measuring how often it would have changed the ordered test.
3. **Year 3, regulated deployment** as a UKCA-marked decision support tool, and a parallel Taiwan pilot at one medical centre with NHI Administration as the reimbursement partner.

The clinical stages inside each year, with who owns each and how success is measured, are in the Medical part, section 6.

14. Development timeline, how fast and what is delivered when

Three years from research prototype to first paid licence. Each quarter has one deliverable that can be checked.

Quarter	Deliverable	Gate to pass
2026 Q4	Hackathon baseline on open data (this submission); first grant application submitted, Cancer Research UK Early Detection and Diagnosis Primer Award	AUC above chance on open data, done (0.82); primer application submitted
2027 Q1	Ethics amendment at the Royal National Orthopaedic Hospital biobank; TCGA full-resolution pipeline with a pathology foundation model	Access letter signed; feature extraction runs on 261 TCGA cases
2027 Q2	First fusion-subtype model (Ewing, synovial, myxoid liposarcoma) on RNOH plus TCGA	Per-subtype AUC reported with confidence intervals
2027 Q3	Robustness gate on the 30-hospital, 3-scanner dataset; stain normalisation on and off	Prediction drift across laboratories below a pre-registered threshold
2027 Q4	Manuscript submitted; Taiwan cohort request through the National Biobank Consortium; shadow-deployment protocol registered	Protocol on a public registry before first case
2028 Q1 to Q2	Shadow deployment at one Genomic Laboratory Hub; DTAC evidence collected	Misroute rate measured with and without the model
2028 Q3 to Q4	Taiwan external validation; NICE Evidence Standards Framework economic case; UKCA technical file drafted	Per-population performance reported
2029 Q1 to Q2	Approved Body audit; MHRA registration; TFDA submission	Certificate issued
2029 Q3 onward	First paid licence to a hub; advisory-mode deployment	Contract signed

15. Ethics and regulation, the path we will follow

What the tool is in law. Software that triages patient data to influence a diagnostic decision is a medical device in the UK, UKCA-marked under the Medical Devices Regulations 2002, with a UK Approved Body audit for Class IIa and above ([MHRA view](#), [clinician's guide](#)). We plan for Class IIa, decision support that a pathologist confirms, never an autonomous result.

Steps, in order.

1. **Research phase, no device.** Retrospective work on open TCGA data and on biobank material under the biobank's existing ethics (RNOH Biobank, HTA licence 12055, REC 15/YH/0311, [RNOH biobank page](#)). No new patient contact, no new consent needed.
2. **Data protection.** A Data Protection Impact Assessment before any NHS slide leaves the laboratory network. Slides stay in the hub, the model is deployed to the data.
3. **NHS entry criteria.** Digital Technology Assessment Criteria, the NHS baseline covering clinical safety (DCB0129 and DCB0160), data protection, technical assurance, interoperability and accessibility ([NICE and DTAC overview](#)).
4. **Evidence standard.** NICE Evidence Standards Framework for digital health technologies, tier C, which asks for comparative evidence of effectiveness and an economic impact case ([NICE ESF](#)). Notebook 02 is the starting point for the economic case.
5. **Shadow deployment** at one Genomic Laboratory Hub, output recorded, not acted on, with a prospective protocol registered before the first case.

6. **UKCA technical file** and Approved Body audit, then a Taiwan TFDA submission in parallel using the same clinical evidence.

Ethics commitments written into the design.

- Performance reported by staining laboratory, scanner, sex, age group and ancestry, never as one pooled number. The 30-hospital dataset exists to make the laboratory split possible.
- A pathologist sees the H&E first and the model second, so automation bias is limited by workflow, not by a warning label.
- The model can abstain. Low-confidence cases are routed to the full panel, not to a guess.
- No patient-level data crosses a border. The Taiwan arm trains and validates inside Taiwan.
- Patient and public involvement before the shadow deployment, with the sarcoma patient charity as the first contact.

16. Patient and public involvement

Patients are involved from the first year, not after the device exists.

1. **A patient advisory panel in year 1**, formed with Sarcoma UK and the Bone Cancer Research Trust, both of which fund and support patient involvement in sarcoma research ([Sarcoma UK](#), [Improving Sarcoma Diagnosis, BCRT for researchers](#)). The panel co-writes the patient-facing explanation of what the tool does and does not do: it does not diagnose, it makes the first genomic order the right one.
2. **What patients feel during the wait is measured, not assumed.** During the year 2 shadow deployment the panel helps design a short patient-reported experience measure on the wait between biopsy and molecular result, so that the days saved in section 4 can be paired with what those days mean to a patient.
3. **The panel reads the fairness reporting.** Performance by sex, age, laboratory, scanner and population (Medical part, section 5) is written up in a plain public summary that the panel approves before it is published.
4. **The panel sits on the advisory board** named in section 17, so the patient voice is in the room when the roll-out decisions in section 13 are made.

This is the patient and public involvement that the NHS Digital Technology Assessment Criteria and the NICE Evidence Standards Framework expect for a tool of this class ([NICE ESF](#)). The partnerships with the two charities are planned, not yet agreed.

17. Team, roles needed and who is in place

Role	Needed for	Status
Clinical lead, orthopaedic and sarcoma	Problem definition, clinical implementation, MDT access	I-Han Cheng, MD, DPhil student NDORMS Oxford, team lead
Machine learning lead	Model, reproducibility, robustness testing	I-Han Cheng for this submission; production model needs a computational pathology partner
Pathology partner	Ground truth, slide access, stain-variation dataset	Planned partner: UCL sarcoma pathology group, contact opened 1 October 2026
Genomics partner	Fusion labels, test-directory knowledge	Planned partner: a Genomic Laboratory Hub
Regulatory and health economics	UKCA file, NICE ESF economic case	Recruited at grant stage
Taiwan clinical partner	Asian validation cohort, NHI pilot	Planned partner: a Taiwan medical centre with NGS accreditation

Contributions for this submission are recorded in `group-member-contact/README.md`.

I-Han Cheng, team lead and sole member. Problem definition, data selection, model design, both notebooks, clinical implementation strategy, business case. Partners approached for the production phase, not authors of this submission: UCL sarcoma pathology group.

The clinical implementation strategy and the business case follow on the next pages.

18. Risks and mitigations

Risk	Likelihood	Mitigation
Model learns stain colour, not biology, and fails at new sites	High, the published sarcoma foundation-model study shows accuracy falling from 0.94 to as low as 0.47 across institutions (Chai and colleagues 2026)	Stain-variation dataset as a gate, stain normalisation, per-site reporting, abstention
Too few open slides for fusion subtypes	Certain today	Start with the two common subtypes, add RNOH and Taiwan biobank material under ethics
Regulatory timeline slips	Medium	Enter as Class IIa decision support, apply to the MHRA AI Airlock phase 3 from April 2026 (GOV.UK AI Airlock)
Hubs do not adopt	Medium	Shadow deployment with a measured misroute rate before any purchase decision
Reimbursement not created in Taiwan	Medium	Pilot under NSTC smart-healthcare funding first, NHI code later